Viswajith E S

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PROFESSIONAL SUMMARY

Senior Data Scientist with 3.8 years of experience building and deploying production ML models for credit risk, fraud, and cross-sell propensity for NBFCs and banks. Delivered multiple underwriting models across different NBFCs that lifted approval rates and cut delinquency rates, using bureau, bank statement, SMS, and GST data. Experienced in model validation, monitoring, and regulator-ready documentation, and in leading end-to-end enterprise delivery with a team of 5 engineers. Also built GenAI and agentic systems for collections and lending operations.

EXPERIENCE

Monsoon CreditTech

End-to-end bespoke ML modelling for 4 NBFCs and 1 private bank

Senior Data Scientist Apr 2025 – Present
Data Scientist Feb 2023 – Mar 2025

- Led a team of 5 engineers on a long-term retainer for a key enterprise client, owning requirements gathering, model delivery, stakeholder management, and production support.

Credit Risk & Underwriting

- Developed and deployed underwriting models across 4 NBFCs using bureau, bank statement, SMS, and application data, improving approval rates by **~10% at constant risk** and reducing delinquency by **~6% at constant approval rates**. Models collectively score **~1.5 million applications monthly in production**.
- Built **a multi-bureau underwriting model** by unifying and normalising data from 3 credit bureaus into a single credit report, gaining 4 AUC points (0.80 to 0.84) over a single-bureau model on out-of-time validation.
- Designed **GST- and ITR-based feature engineering libraries** for MSME and business loan underwriting.
- Built an **MSME model Suite** leveraging Bureau, Banking, GST and ITR data for MSME portfolio.
- Set up **post-deployment monitoring** (PSI/CSI, performance drift) and SHAP-based reason codes for declines; prepared model documentation supporting client validation and RBI model-risk audits.
- Cut feature-selection time from ~2 hours to 10 seconds on large datasets by adding **a Sparse PCA–based feature selection** method to the internal library, with minimal impact on final model performance.

Fraud & Cross-sell

- Built an LGBM-based **insurance fraud model** that captured **78.9%** of fraud cases in the riskiest 3 deciles.
- Built an SMS-based ML model to detect **parallel-funding scams** that don't show up in bureau data, capturing **65.3%** of fraud cases in the riskiest 3 deciles.
- Designed a **personal loan propensity model** for a leading private bank's savings account base, capturing **84.4%** of conversions in the top 30% of ranked customers for targeted cross-sell.
- Built a **customer segmentation pipeline** combining clustering with decision-tree rule extraction to produce interpretable cohorts for targeted marketing.

GenAI & Automation

- Built an RBI-compliant agentic voice system for loan collections that assesses intent to pay and negotiates promise-to-pay, part payment, or escalation to a human agent for restructuring, using an STT–LLM–TTS pipeline with telephony integration.
- Built a modular RAG system over internal knowledge bases and RBI regulations, with recall@k, LLM-as-judge, and gold-dataset evaluations to track every change.

SKILLS

Credit risk & BFSI: Application scorecards, propensity, fraud detection, feature engineering on bureau, bank statement, SMS, GST and ITR data, model validation and monitoring (PSI/CSI), SHAP explainability
Languages & ML: Python, SQL, scikit-learn, LightGBM, CatBoost, PyTorch, Keras, MLflow
GenAI: RAG, LangGraph, LlamaIndex, CrewAI, Hugging Face Transformers
Miscellaneous: FastAPI, Flask, Django, Docker, Streamlit, AWS, GCP, Azure, PySpark, Airflow

EDUCATION

Diploma in Data Science, IIT Madras · CGPA 9.5 Jan 2022 – Oct 2022
Integrated MSc in Physics, UM-DAE CEBS, Mumbai · CGPA 8.07 2013 - 2020

Achievements: KVPY Fellow (2014–2018) · INSPIRE Fellow · IASc Summer Research Fellow · 20-member Indian delegation, Asian Science Camp 2014